



Iron metabolism in doxorubicin-induced cardiotoxicity: From mechanisms to therapies

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ABSTRACT

Doxorubicin (DOX) is an anti-tumor agent for chemotherapy, but its use is often hindered by the severe and life-threatening side effect of cardiovascular toxicity. In recent years, studies have focused on dysregulated iron metabolism and ferroptosis, a unique type of cell death induced by iron overload, as key players driving the development of DOX-induced cardiotoxicity (DIC). Recent advances have demonstrated that DOX disturbs normal cellular iron metabolism, resulting in excessive iron accumulation and ferroptosis in cardiomyocytes. This review will explore how dysregulated iron homeostasis and ferroptosis drive the progression of DIC. We will also discuss the current approaches to target iron metabolism and ferroptosis to mitigate DIC. Besides, we will discuss the limitations and challenges for clinical translation for these therapeutic regimens.

1. Introduction

Doxorubicin (DOX) is a chemotherapeutic agent in treating various types of cancers (Rawat et al., 2021; Sawicki et al., 2021; Yarmohammadi et al., 2021). However, the severe dose-dependent side effects on the heart have significantly restricted its use in clinical settings (Zamorano et al., 2016). The incidence of heart failure increases exponentially with cumulative DOX doses (Swain et al., 2003). DOX-induced cardiotoxicity (DIC) can manifest as either acute or chronic. Acute DIC presents as temporary left ventricular dysfunction, arrhythmias, and abnormal electrocardiograms (Zamorano et al., 2016). On the other hand, chronic DIC is characterized as progressive aggravation of cardiac dysfunction (Kong et al., 2022; Wu et al., 2023). Currently, dexrazoxane is proved in multiple clinical trials to exert cardioprotective effects for DIC treatment. It works through various mechanisms, such as chelating labile iron, reducing oxidative stress, and preventing DNA damage (Deng et al., 2014; Wallace et al., 2020). Other treatment options consist of adjusting DOX regimen, antioxidants, utilization of liposomal formulation, and medications to treat cardiac dysfunction (Wenningmann et al., 2019). However, these treatments may not specifically target the specific mechanisms participating in the development of DIC. Therefore, it is essential to explore the crucial pathogenesis of

DIC and alternative therapeutic approaches.

Iron is a critical microelement for almost all living organisms. Notably, the significance of iron is particularly pronounced in cardiomyocytes for the high energy demands, where its ability to transport electrons makes it a key player in crucial biochemical reactions (Hentze et al., 2004). Specifically, iron is critical for varieties of cardiovascular functions, such as energy production, formation of heme and iron-sulfur (Fe-S) cluster, and enzyme catalysis for biochemical reactions (Gudjoncik et al., 2014). Nonetheless, iron can exert detrimental effects by catalyzing the formation of reactive oxygen species (ROS) via the Fenton reaction (Sawicki et al., 2023). Besides, excessive iron accumulation can trigger ferroptosis, which is a non-apoptotic regulated cell death (Ahola and Langer, 2024). In this context, iron overload in cardiomyocytes can be detrimental to cardiovascular system. During the last 5 years, multiple studies have demonstrated that DOX treatment causes iron deposit in cardiomyocytes, resulting in lipid peroxidation and ferroptosis, as well as the progression of DIC (Fang et al., 2019; Li et al., 2022; Tadokoro et al., 2020). Furthermore, numerous pharmacological compounds have emerged as potential inhibitors for iron overload and ferroptosis, showing promising therapeutic efficacy in mice with DIC (Fang et al., 2019; Li et al., 2022; Tadokoro et al., 2020; Wang et al., 2022; Zhang et al., 2021). Therefore, dysregulation of iron

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metabolism and ferroptosis are tightly related to the onset and development of DIC. Here, we will summarize the evidence and mechanisms for dysregulated iron metabolism and ferroptosis involved in DIC; and we will discuss current advances regarding new treatment approaches targeting these pathophysiological processes for DIC.

2. Cellular iron metabolism and ferroptosis in cardiomyocytes

To maintain the balance of iron levels within cells, a network of proteins plays crucial roles in the cellular iron uptake, storage, release, and export in cardiomyocytes (Fig. 1 and Table 1). Here, the mechanisms of regulation of cellular iron homeostasis and connection to ferroptosis are discussed.

2.1. Iron uptake

The process of iron import is mainly orchestrated via the binding of transferrin (TF)-bound ferric iron to transferrin receptor protein 1/2 (TFR1/2), which undergoes endocytosis in endosomes (Koleini et al., 2021). Then, the ferric iron is reduced to ferrous iron by the STEAP3 (six-transmembrane epithelial antigen of prostate 3) in endosome. The ferrous iron is set free into cytoplasm through the DMT1 (divalent metal

transporter 1) (Luck and Mason, 2012). Notably, deficiency of TFR1 in cardiomyocytes lead to severe cardiomyopathy caused by disrupted generation of Fe/S cluster and mitochondrial injury (Xu et al., 2015). Besides, TFR2 β knockdown attenuates cardiac ischemia/reperfusion injury, which might be linked to enhancement in antioxidative defenses for cardiomyocytes (Roetto et al., 2018). The evidence demonstrates an important role for iron uptake in regulating iron homeostasis and cardiomyocyte function.

In addition to TFR1/2, nontransferrin-bound iron (NTBI) can be imported into cardiomyocytes through DMT1, metal transporters, or calcium channels (L- or T-type) (Oudit et al., 2003; Shah et al., 2022; Shawki et al., 2015). Global knockout of DMT1 in mice leads to premature death within 1 week of birth, indicating the crucial role of DMT1-mediated iron import (Shawki et al., 2015). L-type calcium channels (LTCCs) are mainly expressed in atrial cardiomyocytes and sinoatrial node. The ferrous iron competes with calcium ion for importation into cardiomyocytes via LTCCs in the settings of iron overload conditions (Oudit et al., 2003). T-type calcium channels (TTCCs) are not expressed in adult cardiomyocytes physiologically, whereas TTCCs are present in embryonic hearts and induced in adult cardiomyocytes under pathological conditions, such as cardiac hypertrophy, myocardial infarction, and iron overload (Shah et al., 2022). The SLC39A14 (solute

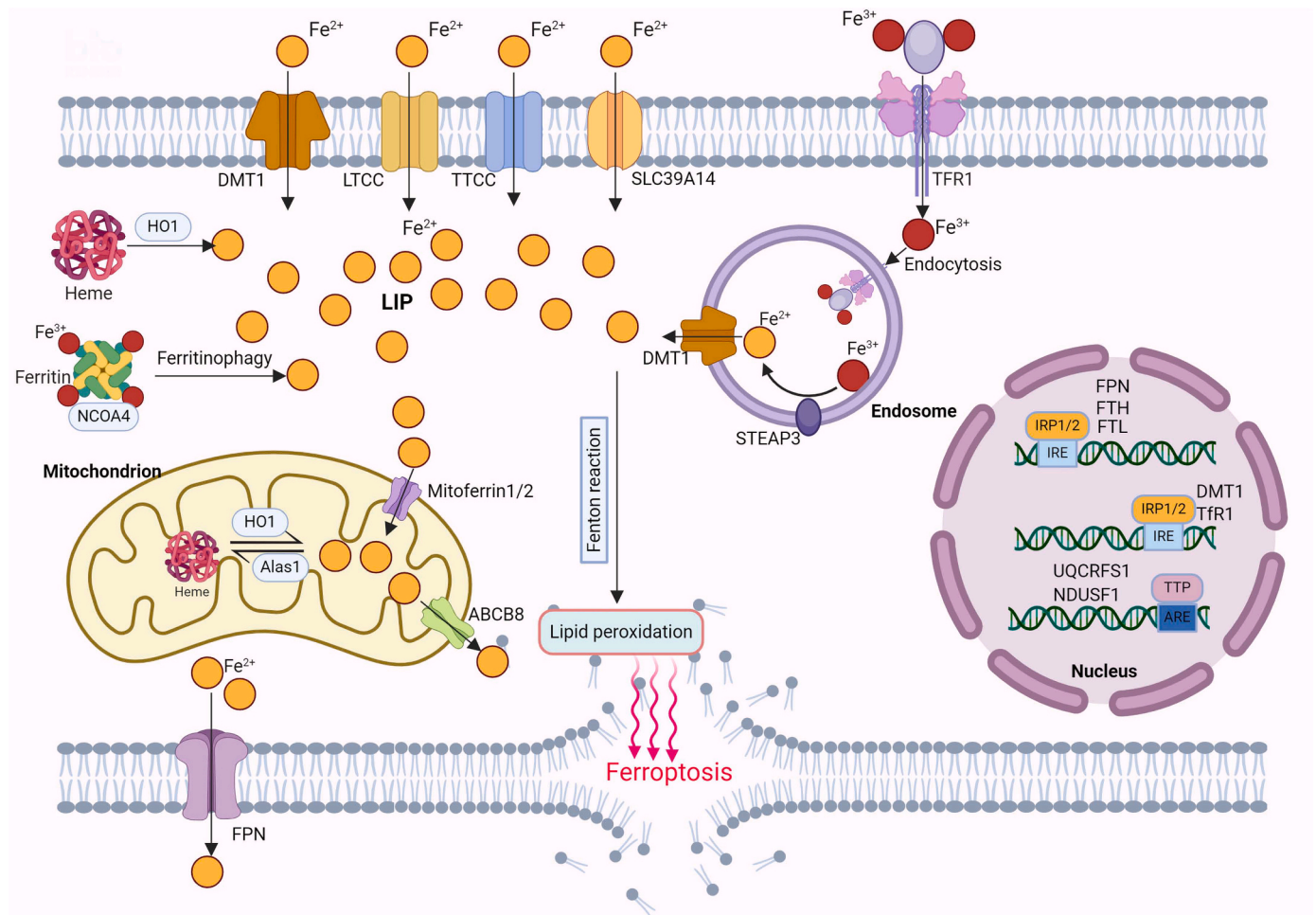


Fig. 1. Molecular mechanisms in the regulation of cardiomyocyte iron homeostasis. The transferrin-bound ferric iron (Fe^{3+}) is transported into cardiomyocytes by transferrin receptor protein 1 (TFR1), while ferrous iron (Fe^{2+}) enters into cardiomyocytes via divalent metal transporter 1 (DMT1), L-type calcium channel (LTCC), T-type calcium channel (TTCC), or metal transporter solute carrier family 39 member 14 (SLC39A14). In addition, the labile iron pool (LIP) can be increased by nuclear receptor coactivator 4 (NCOA4)-mediated ferritinophagy and heme oxygenase 1 (HO1)-mediated heme degradation. Ferroportin (FPN) is the only iron exporter for the transportation of ferrous iron. The increase of LIP promotes lipid peroxidation and ferroptosis through Fenton reaction. Alas1, 5'-aminolevulinatase synthase 1; ARE, AU-rich element; IRE, iron-responsive element; IRP, iron regulatory protein; STEAP3, six-transmembrane epithelial antigen of prostate 3; TTP, tristetraprolin.

Table 1
Cardiac phenotypes of alterations in iron metabolism-related genes.

Gene	Chromosomal location in mice	Protein	Function	Regulation in ferroptosis	Phenotypes of gene overexpression or deletion	Reference
TFR1	Chromosome 16, NC_000082.7	Transferrin receptor protein 1	Iron importer	Promotes ferroptosis	Deletion of TFR1 in cardiomyocytes leads to early death, cardiomegaly, and cardiac dysfunction	(Xu et al., 2015)
TFR2 β	Chromosome 5, NC_000071.7	Transferrin receptor protein 2 β	Iron importer	Promotes ferroptosis	Deletion of TFR2 β ameliorates cardiac ischemia/reperfusion injury	(Boero et al., 2015)
FTH1	Chromosome 19, NC_000085.7	Ferritin heavy chain	Iron storage	Inhibits ferroptosis	Deletion of FTH1 in cardiomyocytes causes iron dysregulation, enhanced oxidative stress, and increased myocardial injury	(Fang et al., 2020)
NCOA4	Chromosome 14, NC_000080.7	Nuclear receptor coactivator 4	Mediates ferritinophagy	Promotes ferroptosis	Deletion of NCOA4 protects against pressure overload-induced iron deposit, ferroptotic cell death, and cardiac dysfunction	(Ito et al., 2021)
FPN	Chromosome 1, NC_000067.7	Ferroportin	Iron exporter	Inhibits ferroptosis	Cardiomyocyte-specific deletion of FPN induces severely impaired cardiac function	(Lakhal-Littleton et al., 2015)
FTMT	Chromosome 18, NC_000084.7	Mitochondrial ferritin	Iron storage in mitochondria	Inhibits ferroptosis	Overexpression of FTMT suppresses ferroptosis induced by erastin	(Wang et al., 2016)
HMOX1	Chromosome 8, NC_000074.7	Heme oxygenase 1	Heme degradation to release Fe ²⁺	Promotes ferroptosis	Inhibition of HO1 inhibits DOX-induced cardiomyocyte ferroptosis	(Fang et al., 2019)

carrier family 39 member 13) is also identified as an NTBI transporting protein for iron uptake (Yu et al., 2020).

2.2. Iron storage and release

To maintain an extremely low iron level and minimize oxidative stress, the labile iron within cells can be stored by complexing with ferritin (Sukhbaatar and Weichhart, 2018). Ferritin-bound iron is released or mobilized via ferritin degradation (ferritinophagy), which is mediated by nuclear receptor coactivator 4 (NCOA4) (Mancias et al., 2014). This process induces the release of labile iron (Quiles Del Rey and Mancias, 2019), which is transported to the cytoplasm via the lysosomal DMT1 (Yambire et al., 2019).

Knockout of ferritin heavy chain (FTH1) in cardiomyocytes results in dysregulation of iron homeostasis, increased ROS generation, and vulnerability to iron overload-related myocardial injury (Fang et al., 2020). Besides, cardiac-specific knockout of NCOA4 attenuates cardiac dysfunction and inhibits ferroptosis in a pressure overload-induced heart failure mouse model (Ito et al., 2021).

2.3. Iron export

To date, ferroportin (FPN) is the only known iron exporter in mammals, which promotes export of intracellular ferrous iron to extracellular space (Ganz, 2005). Notably, FPN deletion in cardiomyocytes contributes to a fatal dilated cardiomyopathy caused by cardiac iron overload (Lakhal-Littleton et al., 2015).

2.4. Iron metabolism and ferroptosis

Ferroptosis is an iron-dependent regulated cell death, which is featured by excessive generation of lipid peroxides (Hirschhorn and Stockwell, 2019). Iron availability is an important element in the initiation of ferroptosis. The imbalance in cellular iron metabolism leads to iron overload and increased cell susceptibility to ferroptosis. Notably, high-iron diet led to overaccumulation of iron and ferroptosis in the liver of mice (Wang et al., 2017). Similarly, *HFE* deficiency in mice protected against hepatic iron overload and attenuated ferroptosis (Wang et al., 2017). Accumulation cellular labile iron contributes to the generation of polyunsaturated fatty acid lipid peroxides and onset of ferroptosis through the Fenton reaction (Ajoolabady et al., 2021). Meanwhile, the conjunction of ferrous iron to ferritin attenuates the Fenton reaction and ferroptosis (Yu et al., 2020). Normally, ferroptotic cell death is mediated by phospholipid peroxides accumulation in the plasma membrane, leading to increased membrane permeability and rupture (Yang et al., 2022). GPX4 (glutathione peroxidase 4) represents the most important

anti-ferroptotic protein to reduce lipid peroxides and increase resistance to ferroptosis (Yang et al., 2022). Multiple investigations show that the loss of GPX4 caused by DOX triggers cardiomyocyte ferroptosis and progression of DIC (Fang et al., 2019; Li et al., 2022; Tadokoro et al., 2020). Therefore, pharmacological activation of GPX4 via selenomethione can effectively attenuated DIC (Huang et al., 2024).

Iron uptake through TFR1 or SLC39A14 can influence the susceptibility to ferroptosis. Besides, ferritin increases cellular resistance to ferroptosis by binding to free iron (Mumbauer et al., 2019; Sun et al., 2016). Thus, ferritinophagy via NCOA4 can enhance the sensitivity to ferroptosis (Hou et al., 2016). Knockdown of iron exporters (e.g., FPN) has been proven to drive the initiation of ferroptosis (Geng et al., 2018). By contrast, deficiency of caeruloplasmin, which assists FPN to leave cells, facilitates transcription factor RSL3-induced and erastin-induced cellular ferroptosis (Shang et al., 2020; Tuo et al., 2017). Overexpression of caeruloplasmin has been shown to inhibit ferroptosis in tumor cells (Shang et al., 2020; Tuo et al., 2017).

In addition to cytoplasmic iron, the homeostasis of mitochondrial iron level also participates in the execution of ferroptosis. Mitoferrin 1/2 are key mitochondrial iron importing proteins governing heme and Fe-S protein biogenesis (Paradkar et al., 2009). A previous study showed that knockout of mitoferrin 2 alleviated erastin-induced ferroptosis, while mitoferrin 2 overexpression promoted ferroptosis (Zhang et al., 2020). Besides, activation of heme oxygenase 1 (HO1), which catalyzes the mitochondrial heme degradation to generate labile iron, leads to excessive accumulation of iron in mitochondria and promotes ferroptosis (Kwon et al., 2015). Similar to cytoplasmic ferritin, mitochondrial ferritin also inhibits the execution of ferroptosis (Wang et al., 2016). Overexpression of mitochondrial ferritin causes increased resistance to erastin-induced ferroptosis (Wang et al., 2016). Taken together, the findings have suggested a critical role of iron availability in ferroptosis.

3. Preclinical and clinical relevance for dysregulated iron metabolism and ferroptosis in DIC

3.1. Iron overload and DIC: preclinical and clinical evidence

Multiple studies have demonstrated that DOX triggers cardiomyocyte iron overload, which contributes to the onset and progression of DIC (Miranda et al., 2003; Panj Rath et al., 2007). A previous in vitro investigation revealed that iron supplementation exacerbated DOX-induced reduced cardiomyocyte viability in a dose-dependent manner (Xu et al., 2008). This phenomenon was also observed in animal DIC models. Interestingly, mice with systemic iron overload, either through a high iron diet or genetic manipulation, were more susceptible to DIC (Miranda et al., 2003; Panj Rath et al., 2007). In line with these

findings, mice with an iron-deficient diet showed a decreased risk of DIC and improved survival rate (Fang et al., 2019). Therefore, attenuation of cardiac iron overload may serve as an effective approach for the management of DIC.

The *HFE* gene is responsible for encoding hereditary hemochromatosis protein, which reduces iron import through binding to TFR1. Mice with *HFE* knockout exhibited an increased cardiac iron level in DIC (Miranda et al., 2003). Consistent with this finding, the genetic mutation in *HFE* H63D has been identified a key indicator to predict high risks to develop DIC (Vaitiekus et al., 2021). Furthermore, in another separate investigation, patients carrying *HFE* C282Y mutation exhibited more severe DIC (Lipshultz et al., 2013). These findings suggest that *HFE* mutations may cause cardiac iron overload and higher susceptibility to DIC. Therefore, *HFE* polymorphism may serve as a valuable biomarker for prediction of a higher risk developing DIC. Taken together, these findings from both preclinical and clinical investigation underscore the involvement of iron overload in the onset and progression of DIC.

3.2. Ferroptosis is involved in the pathogenesis of DIC

Ferroptosis is involved in the etiology of both acute and chronic DIC (Fang et al., 2019; Li et al., 2022; Tadokoro et al., 2020). A previous study investigated the participation of multiple types of cell death in acute DIC by using corresponding inhibitors (Fang et al., 2019). Interestingly, ferrostatin (Fer-1), a ferroptosis inhibitor, dramatically improved mortality in DIC mice (Fang et al., 2019). However, this was not improved by other cell death inhibitor (Fang et al., 2019). DOX caused significant increases in cardiomyocyte iron and ferroptosis (Fang et al., 2019). Furthermore, DOX-induced ferroptosis was caused by induction of release of labile iron via HO1 (Fang et al., 2019). In addition, another independent study also demonstrated the involvement of mitochondria-dependent ferroptosis in DIC (Tadokoro et al., 2020). DOX caused reduced cardiac GPX4 expression, resulting in accumulation of lipid peroxides (Tadokoro et al., 2020). In this context, Mice with GPX4 overexpression and knockdown in exhibited attenuated and accentuated cardiac dysfunction in response to DOX treatment, respectively (Tadokoro et al., 2020). Mechanistically, DOX induced excessive iron accumulation and inactivation of GSH/GPX4 pathway in cardiac tissues (Li et al., 2022). Taken together, the evidence indicates that ferroptosis might persist continuously and contribute to the development of DIC.

4. Molecular mechanisms of iron overload in DIC

4.1. DOX increases iron uptake and inhibits iron export

DOX has been shown to affect the levels of iron metabolism-related genes (e.g., iron uptake, storage, export) in cardiomyocytes (Kotamraju et al., 2002; Mouli et al., 2015; Pan et al., 2023; Zhuang et al., 2021). Increased TFR1 and FTH1 levels, and decreased FPN abundance are observed in hearts or cardiomyocytes in the face of DOX insult (Kotamraju et al., 2002; Mouli et al., 2015; Pan et al., 2023; Zhuang et al., 2021). A previous study indicated an important role for TFR1-mediated iron uptake in DIC (Kotamraju et al., 2002). Inhibition of TFR1 using the specific blocking antibody was shown to suppress iron uptake, oxidative stress, and cell death in DOX-treated endothelial cells (Kotamraju et al., 2002).

Several studies have postulated mechanisms regarding how DOX triggers TFR1-mediated iron uptake. A previous study showed that DOX caused increased expression of METTL14, thereby enhancing the m⁶A modification and stability of KCNQ1OT1 (Zhuang et al., 2021). KCNQ1OT1 was shown to silence miR-7-5p and induce an upregulation of TFR1 (Zhuang et al., 2021). Our recent study also showed that DOX upregulated METTL3, a major m⁶A methyltransferase, in cardiomyocytes (Wu et al., 2024). METTL3 upregulated the m⁶A modification of TFR1, which facilitates TFR1 mRNA stability through binding to IGF2BP2 (Wu et al., 2024). In this context, STM2457, a highly selective

METTL3 inhibitor, effectively suppressed cardiomyocyte ferroptosis and progression of DIC (Wu et al., 2024). Moreover, a recent study has shown that sorting nexin 3 (SNX3) promotes the recycling of TFR1, contributing to the iron accumulation and ferroptosis, and eventually leading to the progression of DIC (Yu et al., 2023). Currently, it remains unclear how DOX downregulated FPN and inhibits iron export. Taken together, DOX triggers cardiac iron accumulation and ferroptosis by facilitating iron uptake and reducing iron export.

4.2. DOX impairs heme metabolism

Heme is synthesized in mitochondria and this process transforms labile iron into heme-containing iron. DOX can trigger heme degradation to increase labile iron, and also decreases heme biosynthesis to utilize free iron, resulting in iron accumulation (Abe et al., 2022; Fang et al., 2019; Zhang et al., 2021). Fang and coworkers showed that HO1 expression was upregulated via Nrf2 activation in a murine DIC model (Fang et al., 2019). HO1 mediates heme degradation and increase of labile iron. Furthermore, another recent study demonstrated that DOX caused downregulation in Alas1 (5-aminolevulinate synthase 1), which is a critical enzyme for heme biosynthesis (Abe et al., 2022), resulting in impairment in cardiac iron utilization, and subsequent cardiomyocyte ferroptosis (Abe et al., 2022). Overexpression of Alas1 or treatment with 5-aminolevulinic acid significantly attenuated iron overload and ferroptosis in DIC (Abe et al., 2022). Taken together, the evidence has suggested critical roles of impaired heme synthesis and increased heme degradation in iron accumulation, ferroptosis, and development of DIC.

4.3. DOX triggers iron accumulation in the mitochondria

Except for disturbed heme metabolism, excessive mitochondrial iron accumulation serves as an important mechanism for DIC. Mitochondrial iron overload was observed in heart samples from end-stage DIC patients, compared to those from healthy controls or non-DOX-caused cardiomyopathy patients (Ichikawa et al., 2014). Mechanistically, DOX-induced mitochondrial iron overload was caused by downregulation of ABCB8, an iron exporting protein in mitochondria (Ichikawa et al., 2014). In this context, overexpression of ABCB8 or chelating mitochondrial iron through dexrazoxane significantly attenuated DIC progression (Ichikawa et al., 2014). Besides, DOX also increased the expression of mitochondrial ferritin (FTMT) in cardiomyocytes, and knockout of FTMT in cardiomyocytes was more vulnerable to DIC and associated ferroptosis (Maccarinelli et al., 2014). Mitochondrial iron overload is important in mitochondria-dependent ferroptosis and progression of DIC. Taken together, Fig. 2 demonstrates the mechanisms regarding how DOX triggers cardiomyocyte iron overload and associated ferroptosis.

5. Targeting dysregulated iron homeostasis in the treatment of DIC

Having established the pathophysiology of dysregulated iron metabolism and associated ferroptosis in DIC, attenuating iron overload might represent a potential therapeutic approach for DIC. Here, we mainly discuss the current findings regarding drugs or natural compounds to ameliorate iron accumulation (iron chelators) and associated ferroptosis in DIC (Table 2).

5.1. Iron chelators

The iron chelator dexrazoxane has been approved for the treatment with a cumulative DOX dose >300 or >350 mg/m² in Europe and America, respectively (Sawicki et al., 2021). Suppression of ferroptosis serves as important mechanism of action for dexrazoxane-mediated protective effects in DIC (Fang et al., 2019). Mechanistically, dexrazoxane regulates the expression of iron metabolism-related genes, as

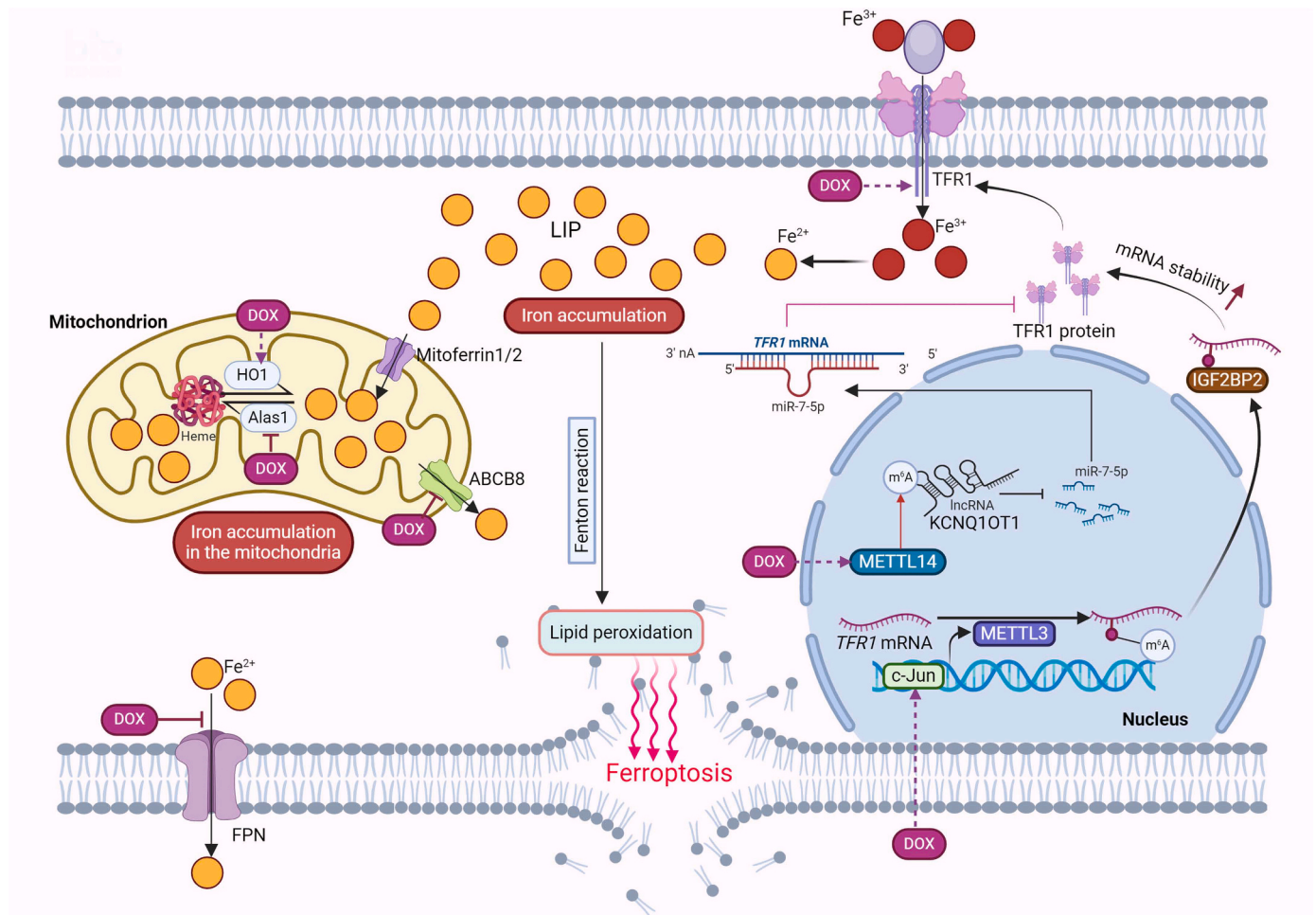


Fig. 2. Dysregulated iron homeostasis in cardiomyocytes contributes to DOX-induced ferroptosis. DOX causes iron accumulation by increasing TFR1-mediated iron uptake, and decreasing FPN-mediated iron export. Specifically, DOX upregulates METTL3, which promotes m⁶A modification of TFR1 mRNA and increases its stability. Besides, DOX also increases the expression of METTL14, facilitating the m⁶A modification and stability of KCNQ10T1. The KCNQ10T1 can suppress miR-7-5p, causing the upregulation of TFR1. DOX can suppress ABCB8 (a mitochondrial iron exporting protein) and disturb heme metabolism, resulting in excessive mitochondrial iron accumulation. Alas1, 5'-aminolevulinate synthase 1; FPN, ferroportin; HO1, heme oxygenase 1; LIP, labile iron pool; m⁶A, N⁶-methyladenosine; TFR1, transferrin receptor protein 1.

well as downregulated the heme degradation enzyme HO1 in cardiomyocytes (Jirkovsky et al., 2013). In this context, dexrazoxane can prevent heme degradation-associated release of labile iron. Besides, HMGB1 may serve as a potential downstream for protective effects of dexrazoxane (Zhang et al., 2021). DOX was shown to upregulate HMGB1 and inhibition of HMGB1 significantly suppressed cardiomyocyte ferroptosis (Zhang et al., 2021). Dexrazoxane ameliorated cardiomyocyte ferroptosis and DIC by downregulating HMGB1 (Zhang et al., 2021). Except for attenuation of iron overload and cardiomyocyte ferroptosis, it is noteworthy that dexrazoxane may exert other effects against DIC, such as inhibition of oxidative stress DNA damage (Deng et al., 2014; Wallace et al., 2020).

Despite clinical effectiveness of dexrazoxane in DIC, several limitations remain for its further application. The major challenge is the possibility of increased risk of secondary cancers and hematologic neoplasms (Seif et al., 2015; Shaikh et al., 2016; Tebbi et al., 2007). Currently, the application of dexrazoxane is strictly restricted in metastatic breast cancer patients who receive cumulative doses >300 mg/m² with DOX (Wu et al., 2023).

Currently, the therapeutic role of DFO in DIC is still controversial. Multiple preclinical studies have demonstrated that deferoxamine (DFO), which chelates excessive cellular ferrous iron, can attenuate cardiomyocyte ferroptosis and DIC (Al-Shabanah et al., 2012;

Fedotcheva and Fedotcheva, 2021; Hanna et al., 2022; Pan et al., 2023; Rahimi et al., 2023; Saad et al., 2001). However, an earlier study demonstrated that DFO could not exert effective cardioprotective effects against DIC, while dexrazoxane could ameliorate mitochondrial iron accumulation and inhibit progression of DIC (Ichikawa et al., 2014). The possible explanation is that DFO fails to enter the mitochondria, while dexrazoxane easily locate at mitochondria to reduce the accumulation of mitochondrial iron and oxidative stress. In line with finding, Mito-FerroGreen, which decreases ferrous iron through conversion to ferric iron, significantly attenuated cardiomyocyte ferroptosis and accumulation of lipid peroxides in DIC (Tadokoro et al., 2020). Furthermore, there is a narrow window for DFO dosage in the application, and a slight deviation of DFO dose might diminish the cardioprotective effects (Voest et al., 1994). More researches are warranted to figure out the clinical efficacy and specific dosage of DFO in the management of DIC.

5.2. Other ferroptosis inhibitors

Cardiomyocyte iron overload causes ferroptosis, which has been involved in the pathogenesis of DOX-induced cardiac dysfunction (Fang et al., 2019; Li et al., 2022; Tadokoro et al., 2020). Fer-1 is a radical-trapping antioxidant to reduce excessive generation of lipid ROS,

Table 2
Iron chelators and ferroptosis inhibitors in the treatment of DIC in preclinical models.

Therapeutic target	Drug or compound	Mechanism	Effects	Reference
Iron chelators or iron regulators	Dextrazoxane	Iron chelation	Improves cardiac function and mitochondrial function Reduces lipid ROS	(Fang et al., 2019; Pan et al., 2023; Zhang et al., 2021)
	DFO	Iron chelation	Improves cardiac function and mitochondrial function Reduces lipid ROS	(Al-Shabanah et al., 2012; Hanna et al., 2022; Pan et al., 2023; Rahimi et al., 2023; Saad et al., 2001)
Ferroptosis inhibitors	MFG	Decreases ferrous iron in the mitochondria through conversion to ferric iron	Reduces lipid ROS Increases Cell viability	(Tadokoro et al., 2020)
	Zinc protoporphyrin IX	Suppresses heme degradation to decrease iron release	Reduces cardiac remodeling and lipid peroxides	(Fang et al., 2019)
	5-aminolevulinic acid	Increases heme biosynthesis to facilitate iron utilization	Attenuates cardiac dysfunction	(Abe et al., 2022)
	Fer-1	Inhibits lipid peroxidation	Reduces myocardial damage Attenuates cardiac dysfunction	(Fang et al., 2019; Liu et al., 2020; Pan et al., 2023; Tadokoro et al., 2020)
	MitoTEMPO	Scavenges mitochondrial ROS	Reduces lipid peroxides and ferroptosis Reduces cardiac injury, remodeling, and lipid ROS	(Fang et al., 2019)
	Selenomethione	Activates GPX4	Attenuates cardiac dysfunction and exerted antitumor effects simultaneously	(Huang et al., 2024)
	WGX50	Decreases iron overload and mitochondrial ROS	Attenuates cardiac dysfunction	(Tai et al., 2023)
	Ethoxyquin	Decreases lipid peroxides	Reduces myocardial injury Attenuates cardiac dysfunction	(Tadokoro et al., 2022)
	Resveratrol	Activates Nrf2 signaling pathway	Reduces myocardial injury, remodeling, and lipid peroxidation Attenuates cardiac dysfunction, lipid peroxidation, and ferroptosis	(Yu et al., 2022)

and has been proven to protect against DIC (Fang et al., 2019; Liu et al., 2020; Pan et al., 2023; Tadokoro et al., 2020; Wang et al., 2022; Zhang et al., 2021). Fer-1 reduced myocardial injury without influencing iron levels in the heart in animal models of DIC (Fang et al., 2019). Fer-1 was also been shown to decrease cardiomyocyte cell death in the face of DOX treatment (Tadokoro et al., 2020). However, a major challenge for the application is the low stability of Fer-1 in plasma, resulting in decreased efficacy and clinical applicability (Li et al., 2019; Sun et al., 2022). Future studies are required to identify effective ferroptosis biomarkers in DIC to evaluate the efficacy in clinical settings.

In addition to Fer-1, multiple drugs or compounds are shown to inhibit cardiomyocyte ferroptosis and protect against DIC. MitoTEMPO, which scavenges mitochondrial ROS (Dikalova et al., 2010), has been demonstrated to suppress ferroptosis and lipid peroxidation, and attenuate DIC progression in mice (Fang et al., 2019). Ethoxyquin, another radical-trapping antioxidant, significant attenuated cardiomyocyte ferroptosis and DIC (Tadokoro et al., 2022). N-acetyl cysteine (NAC) is a classic ferroptosis inhibitor, and have been shown to improve cardiovascular function in DIC (Doroshov et al., 1981; Goyal et al., 2016). Furthermore, several natural compounds have been shown to exert anti-ferroptotic effects by activating Nrf2 signaling and attenuate the development of DIC, such as resveratrol, fisetin, and astragaloside IV (Li et al., 2021; Lu et al., 2022; Luo et al., 2021; Yu et al., 2022). Collectively, pharmacological inhibition of cardiac ferroptosis is an effective therapeutic approach for DIC.

6. Conclusion

Here, we conclude that iron metabolism and associated ferroptosis play important roles in the pathogenesis of DIC. Despite this progress, there are still numerous unanswered questions that need to be addressed before we can consider the use of iron chelators or anti-ferroptotic drugs in clinical settings. Firstly, while iron chelators attenuate cardiac dysfunction in DIC, they may enhance the risk of tumor proliferation. As mentioned earlier, dextrazoxane is related to increased risks of secondary tumors in children (Seif et al., 2015; Shaikh et al., 2016; Tebbi et al., 2007), possibly resulted by inhibition of topoisomerase II α in cancer

cells (Deng et al., 2014). To the best of our knowledge, there is a lack of investigation investigating the effects of iron chelation in DIC models with concurrent cancers. Besides, while iron chelators and ferroptosis inhibitors have been widely studied in preclinical DIC models, clinical evidence (except dextrazoxane) is still lacking. Lastly, biomarkers for ferroptosis (e.g., 4-HNE) are not specific, and might observed in other cell death types. It is important to discover more specific ferroptotic biomarkers in DIC.

To sum up, Dysregulated iron metabolism is involved in cardiomyocyte ferroptosis and progression of DIC. Suppression of iron accumulation and associated ferroptosis provides cardioprotective effects in DIC treatment in both clinical and preclinical settings. Future investigations of dysregulated iron homeostasis in DIC will concentrate on the potential reasons of iron accumulation for the progression of DIC. Besides, discovering downstream mechanism of iron overload that cause cardiac dysfunction, might provide underlying therapeutic targets for DIC.

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CRediT authorship contribution statement

Hua Ye: Writing – review & editing, Writing – original draft, Conceptualization. Lin Wu: Writing – review & editing, Writing – original draft, Conceptualization. Yanmei Liu: Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

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Author contributions

HY, LW, and YL drafted and revised the manuscript.

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